

Proposed Solution

OptiCrop — Smart Agricultural Production Optimization Engine

Overview

OptiCrop is a machine-learning powered web application that recommends the most suitable crop for a given set of soil and climate conditions, helping farmers, researchers, and policymakers make data-driven agricultural decisions.

How It Works

- User enters Nitrogen (N), Phosphorous (P), Potassium (K), temperature, humidity, pH, and rainfall through a web form.
- Inputs are scaled using a StandardScaler fitted during training.
- The trained Random Forest model predicts the most suitable crop class.
- The predicted label is decoded back into a human-readable crop name and displayed to the user.

Key Differentiators

- Compares multiple algorithms rather than relying on a single untested model.
- Built on a clean, balanced, well-documented public dataset (22 crops, 100 samples each).
- Simple three-page web interface usable without any ML background.
- Fully reproducible — notebook, dataset, and trained model artifacts are all included in the repository.

Target Users

- Farmers — get a direct crop recommendation for their land's conditions.
- Agricultural researchers — study crop-environment relationships using the same underlying model and dataset.
- Policymakers — use aggregate insights to inform resource-allocation and sustainability strategy.